

ASTR 5140: Final Project

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ABSTRACT

We propose near-infrared spectroscopic observations of low-mass stars in the low-metallicity cluster S284-EC1 to test whether the initial mass function (IMF) depends on metallicity. Recent JWST photometric studies suggest that the IMF peak in this cluster occurs at lower mass than in solar-metallicity environments, but these results are limited by degeneracies between stellar mass, extinction, and age in color-magnitude diagram (CMD) fitting. We will obtain JWST/NIRSpec spectroscopy of ~ 30 – 50 stars near the IMF peak (0.1 – $0.3 M_{\odot}$) to independently constrain stellar temperatures and extinctions. JWST/NIRSpec is uniquely suited to this program due to its sensitivity to faint, embedded low-mass stars at near-infrared wavelengths and its multi-object spectroscopy capability, which enables efficient observation of multiple cluster members in a single pointing. By combining spectroscopic measurements with existing photometry, we will derive improved stellar masses and test whether the observed IMF shift is robust. This program directly addresses the question of IMF universality and the role of metallicity in star formation.

1. SCIENTIFIC JUSTIFICATION

The distribution of stellar masses at birth, known as the initial mass function (IMF), is one of the most fundamental quantities in astrophysics. It determines how many low-mass stars form compared to high-mass stars, and therefore influences everything from star cluster evolution to the chemical enrichment of galaxies. A long-standing question is whether the IMF is universal, or whether it depends on environmental properties such as metallicity.

Metallicity refers to the abundance of elements heavier than hydrogen and helium. These elements play an important role in the cooling of gas in star-forming regions. Because cooling affects how gas fragments during collapse, theoretical models predict that metallicity may influence the typical mass scale of stars that form. However, observational evidence for this effect remains uncertain.

Recent work by [Andersen et al. \(2025\)](#) studied the IMF in the low-metallicity cluster S284-EC1 using deep infrared imaging from the James Webb Space Telescope (JWST). They found that the IMF peak occurs at a lower mass than is typically observed in solar-metallicity environments. This suggests that metallicity may influence the characteristic stellar mass. However, this result depends on how stellar masses are estimated from photometric data, and is sensitive to assumptions about extinction, age, and stellar evolution models.

The key limitation of photometric studies is that multiple physical effects can shift a star’s position in a color-magnitude diagram (CMD), as illustrated in [Figure 1](#). In particular, stellar mass, extinction from dust, and age can produce similar shifts in observed color and brightness. This degeneracy makes it difficult to uniquely determine stellar masses using photometry alone.

Spectroscopy provides an independent way to constrain stellar properties. By analyzing spectral features, it is possible to estimate a star’s effective temperature (spectral type), surface gravity, and youth indicators. These measurements allow for a more direct estimate of stellar mass, and can help break degeneracies present in CMD fitting.

While spectroscopic observations of S284 do exist, they are limited to optical wavelengths and relatively bright stars ([Kalari & Vink 2015](#)). These optical spectroscopic data are also strongly affected by dust extinction and primarily probe stars with masses above $\sim 1 M_{\odot}$, well above the mass range where the IMF peak occurs. The low-mass stars that dominate the IMF peak (0.1 – $0.3 M_{\odot}$) are faint and embedded, and are best observed at near-infrared wavelengths, where extinction effects are significantly reduced.

As a result, there is currently no spectroscopic dataset that directly constrains the low-mass stellar population observed by JWST in S284-EC1. This creates a critical gap between photometric IMF measurements and spectroscopic constraints.

In this proposal, we aim to obtain near-infrared spectroscopy of low-mass members of S284-EC1 to independently estimate stellar temperatures, extinctions, and youth indicators. By combining spectroscopic constraints with existing JWST photometry, we will refine the derived mass distribution and test whether the observed shift in the IMF peak is a real environmental effect or a consequence of photometric degeneracies.

2. EXPERIMENTAL DESIGN

The goal of this project is to obtain independent constraints on stellar masses in S284-EC1 by supplementing existing photometric data with spectroscopic measurements. This will allow us to reduce uncertainties associated with extinction, age, and model assumptions in the IMF derivation.

We propose to obtain new near-infrared spectroscopic observations of candidate cluster members using JWST/NIRSpec in multi-object spectroscopy mode. Searches of the MAST archive indicate that while JWST/NIRSpec observations exist in the direction of S284, these are limited to calibration exposures (e.g., dark frames) and do not contain science spectroscopy of cluster members. To our knowledge, no near-infrared spectroscopic dataset currently probes the low-mass stellar population of S284-EC1 at the depth required to test the IMF peak.

Target selection will be based on the JWST photometric catalog from [Andersen et al. \(2025\)](#). We will prioritize stars near the inferred IMF peak (approximately $0.1\text{--}0.3 M_{\odot}$), as well as a smaller sample of higher-mass stars for calibration. This mass range corresponds to the peak of the IMF, where uncertainties have the largest impact on the derived mass distribution (Figure 2).

From the spectra, we will measure temperature-sensitive features such as molecular absorption bands (e.g., H_2O and CO), which are prominent in low-mass stars. These features can be used to estimate effective temperature and spectral type. Combined with luminosity estimates from photometry, this allows us to place stars on theoretical evolutionary tracks and infer their masses.

We will also use spectroscopic indicators to estimate extinction independently of CMD position. This provides a critical check on the photometric extinction estimates used in previous work.

The analysis will proceed in three main steps:

1. Derive spectral types and effective temperatures from near-infrared spectra.
2. Combine spectroscopic temperatures with photometric luminosities to estimate stellar masses.
3. Reconstruct the IMF using spectroscopically calibrated masses and compare to the original photometric IMF.

We estimate that spectroscopy of $\sim 30\text{--}50$ low-mass cluster members can be obtained using JWST/NIRSpec in multi-object mode. A detailed exposure time calculation would be required for a formal JWST submission, but as an order-of-magnitude estimate, exposures of ~ 1 hour per pointing should provide sufficient signal-to-noise for broad spectral classification of targets near the IMF peak. Including overheads and multiple pointings or microshutter configurations, this corresponds to a total observing time of approximately 5–8 hours.

By comparing the IMF derived from spectroscopy with the photometric IMF, we will assess whether the observed shift in the IMF peak is robust or driven by degeneracies in CMD fitting.

Expected Outcomes and Impact:

If the spectroscopically derived IMF confirms the lower peak mass reported by [Andersen et al. \(2025\)](#), this would provide strong evidence that the IMF depends on metallicity, with important implications for star formation theories and galaxy evolution models. Conversely, if the IMF peak shifts toward higher masses when spectroscopic constraints are applied, this would indicate that photometric degeneracies significantly bias current IMF measurements.

In either case, this study will provide a critical test of IMF universality by directly addressing the dominant source of uncertainty in photometric mass estimates. The results will also establish a framework for combining JWST photometry and spectroscopy to study low-mass stellar populations in other environments.

3. FIGURES

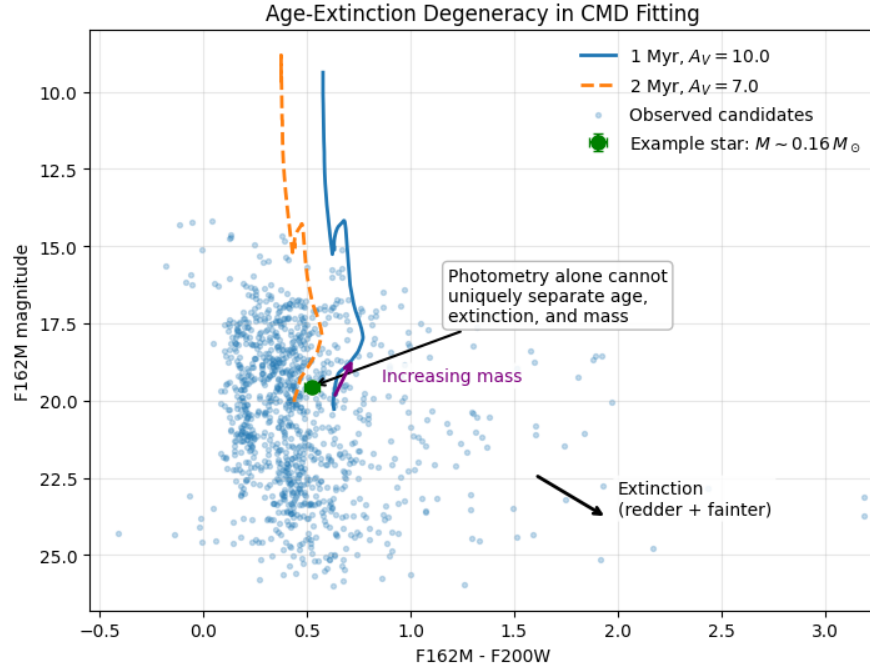


Figure 1. Color-magnitude diagram (CMD) of S284-EC1 showing isochrones at different ages and extinction values. A single observed star can be reproduced by multiple combinations of stellar mass, age, and extinction, illustrating the degeneracy inherent in photometric mass estimation.

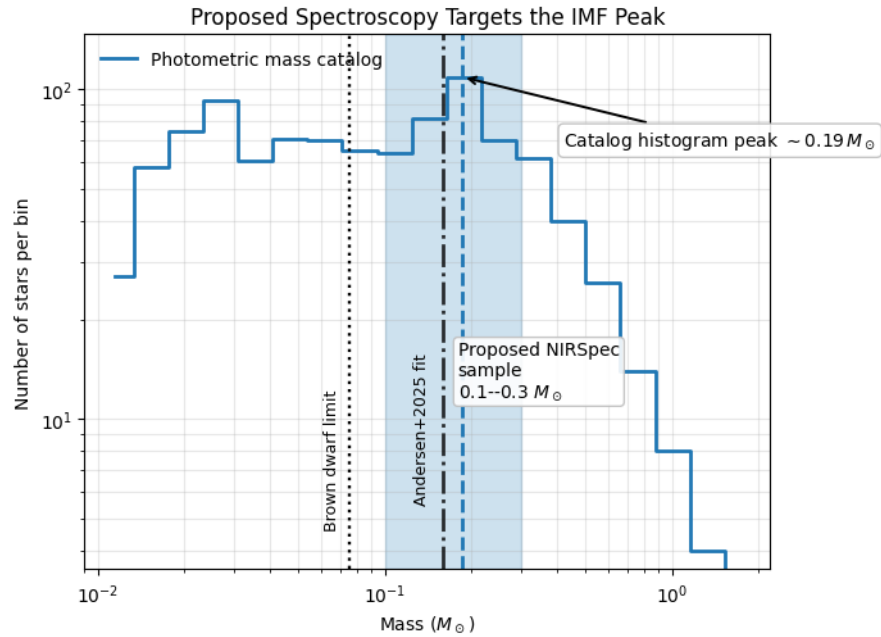


Figure 2. Photometrically derived mass distribution of S284-EC1. The dashed line marks the histogram peak, while the dash-dotted line indicates the IMF peak reported by Andersen et al. (2025). The shaded region highlights the proposed spectroscopic target range ($0.1\text{--}0.3 M_\odot$).

REFERENCES

Andersen, M., et al. 2025, arXiv e-prints,
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